

A blockchain-based digital product passport can help footwear manufacturers demonstrate compliance with ecodesign sustainability criteria across their value chain.

Manufacturing × Regulatory compliance and reporting × Blockchain and DLT · OS115 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
51 is the world moving	53 can we play, can we win	MEDIUM 0 named in the evidence	€5.0bn partial evidence · per year	NEXT regulation adopted or propose...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

A blockchain-based digital product passport can help footwear manufacturers demonstrate compliance with ecodesign sustainability criteria across their value chain. This opportunity is for European footwear manufacturers facing upcoming regulatory requirements. It exists now because an EU-funded project is developing an advanced ecodesign methodology for the footwear sector, and DPPs are emerging as core infrastructure for sustainable manufacturing.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€7.0bn €885m – €45.6bn	€5.0bn €635m – €32.7bn	€60.0m €7.6m – €393m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€251k €251k – €679k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Blockchain and DLT	52.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

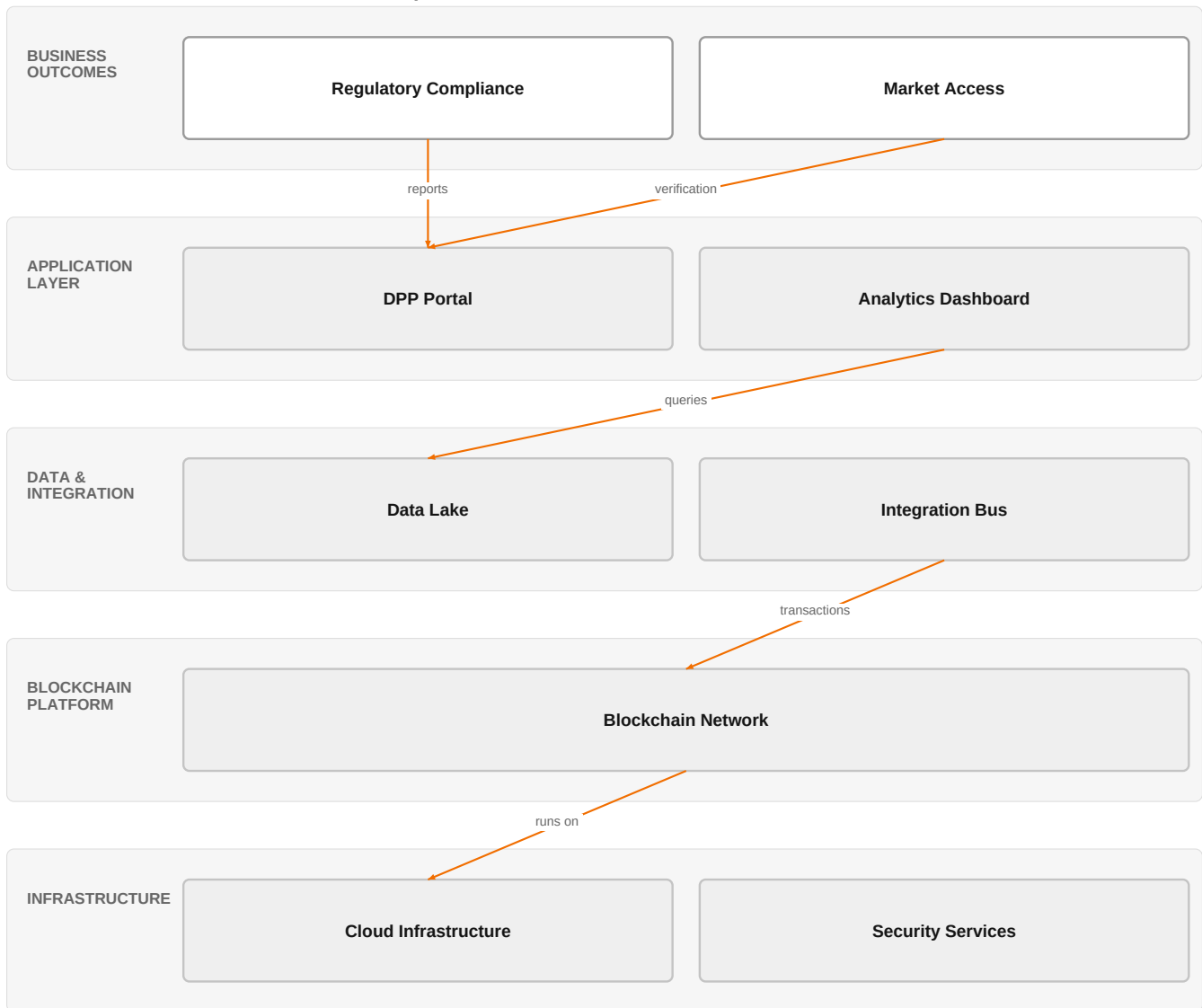
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Blockchain-based DPP for Footwear Compliance



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's cloud and security assets, combined with a blockchain platform, enable a DPP solution that supports regulatory compliance and market access.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The European Green Deal is driving the adoption of Digital Product Passports (DPPs) as a critical tool for sustainable supply chains, improving traceability, transparency, reliability, and circularity. An EU-funded project (ECOSTEP) is developing an advanced ecodesign methodology for the footwear sector, addressing circularity, durability, and social sustainability across the full value chain. DPPs are rapidly emerging as core infrastructure for sustainable manufacturing, enabling lifecycle transparency, circular economy practices, and regulatory compliance. However, full integration of DPPs is hampered by various organizational hurdles, indicating a market need for solutions that overcome these challenges.

Sources: SIG-E4163170FA87 Systems 2026-06-17; SIG-E53BD5BBDBAD cordis.europa.eu 2026-06-01; SIG-36FFB2435F8A openalex.org 2026-04-20

Who buys, and why now

The buyer is typically the Chief Sustainability Officer or Head of Compliance at a European footwear manufacturer. The pain is felt by supply chain and IT teams who must collect and verify sustainability data from multiple tiers of suppliers. The trigger is the upcoming regulatory requirement to provide product-level sustainability data, forcing them to act now to

avoid compliance gaps and potential market access issues.

Why it is hot now — every claim bound to a dated source

- An EU-funded project is developing an advanced ecodesign methodology for the footwear sector, addressing circularity, durability, and social sustainability across the full value chain. [SIG-E53BD5BBDBAD]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a consulting and integration engagement leveraging its AI/Data/Cloud experts and Digital experts to design and implement a blockchain-based DPP solution. The engagement would start with a discovery phase to map the customer's value chain and data sources, followed by architecture design using Orange's expertise in cloud and data. Implementation would include integrating with existing systems and deploying the DPP platform, with ongoing managed services from Orange Cyberdefense for security and compliance. The solution would be hosted on a sovereign cloud (potentially via a partner) to meet trust requirements.

Why Orange can win

Orange can win by combining its certified infrastructure (ISO 27001, TISAX) with its AI/Data/Cloud expertise and digital consulting capabilities. The ability to provide a secure, sovereign solution is a differentiator, especially for European manufacturers concerned about data sovereignty. However, the assets are thin in terms of specific DPP or blockchain references, so Orange must leverage its broader manufacturing references (e.g., Saint-Gobain Glass) to build credibility.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 3.0; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Accenture	Global systems integrator	active in Manufacturing; competes in Cloud	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in Cloud	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in Cloud	structural	—

Vodafone Business	Telecom operator peer	active in Manufacturing	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Manufacturing; competes in Cloud	structural	—
Honeywell	Industrial / OT specialist	active in Manufacturing	structural	—
Schneider Electric	Industrial / OT specialist	active in Manufacturing	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current process for collecting and verifying sustainability data from your suppliers?
- 2 Have you started preparing for the upcoming Digital Product Passport regulations? If so, what gaps have you identified?
- 3 Which product lines or materials are most challenging to trace across your value chain?
- 4 Do you have a data governance framework in place to manage product-level data across multiple tiers?
- 5 What is your timeline for achieving compliance with ecodesign criteria?
- 6 Who in your organization would be the primary decision-maker for a DPP initiative?

Objections you will meet

They will say	You can answer
We already have a sustainability reporting system in place.	That's true, and it likely covers your current reporting needs. However, DPPs require granular, product-level data that is verifiable and shareable across the value chain, which most existing systems do not provide. Our solution can complement your existing systems by adding the traceability layer.
Blockchain is overhyped and not proven for this use case.	You're right that blockchain has been overhyped in some areas. However, for DPPs, blockchain provides an immutable, decentralized record that is essential for trust among multiple stakeholders. We can start with a pilot to demonstrate its value in your specific context.
We are concerned about the cost and complexity of implementation.	Cost and complexity are valid concerns. Our approach is phased, starting with a discovery and architecture phase to minimize risk. We also leverage our existing cloud and security assets to reduce infrastructure costs. The long-term benefits of compliance and market access often outweigh the initial investment.

Risks and unknowns

The main risk is regulatory uncertainty: the DPP regulation is proposed but not yet applicable, so the exact requirements may change. There is also a lack of proven blockchain-based DPP implementations in footwear, making it a nascent market. Additionally, the organizational hurdles in supply chain integration are significant and may slow adoption.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: develop a proposal for a footwear manufacturer that combines Orange Cyberdefense managed services with blockchain-based digital product passports, using the EU-funded ecodesign methodology as the compliance framework, and reference our work with Saint-Gobain Glass to illustrate cross-industry expertise.
Sales	In a meeting with a footwear manufacturer's sustainability or IT leader, say: 'We can help you demonstrate ecodesign compliance across your value chain by leveraging our ISO 27001-certified managed security services to protect the data integrity of your digital product passports.'
Strategist / Innovator	Deep-dive brief: map how the EU-funded ecodesign methodology for footwear aligns with Orange Cyberdefense managed services and ISO 27001 to position blockchain-based digital product passports as a secure compliance solution.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E4163170FA87	2026-06-17	Systems	1	Unlocking Digital Product Passport Integration: Multidimensional Hurdles in Supply Chains
SIG-E53BD5BBDBAD	2026-06-01	cordis.europa.eu	1	ECODESIGN criteria for Sustainability in footwear and Testing methods for Evaluation of Perform...
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-FE1FBF23480C	2026-02-01	openalex.org	1	Digital product passports for cleaner production: Economic evidence from producers and consumer...
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-8123897AE6E6	2025-12-15	openalex.org	1	Bridging digital tools and human skills: A sustainable competence framework for implementing di...
SIG-06EF21FC3A70	2025-12-04	openalex.org	1	Consumer Acceptance of Digital Product Passports: The Roles of Technological Awareness and Valu...
SIG-EFEBD520406D	2025-10-15	openalex.org	1	Stakeholder perspectives on digital product passports for construction products
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-BCCFFFF0F9D1	2025-09-30	openalex.org	1	Textile Materials Information for Digital Product Passport Implementation in the Textile and Cl...
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-99F736E24D27	2026-05-04	Deloitte	2	Digital Product Passport (DPP): Unlocking Transparency and Sustainability in Manufacturing, R&D...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (6 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:09+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.